

02 — Database Schema

SmartFix uses **three persistence layers** and **one in-memory layer**. Each is documented below with shape, ownership, and durability characteristics.

Storage overview

Layer	Purpose	Survives restart?	Backed up?
ChromaDB (<code>./chroma_db/</code>)	Vector index of manual chunks	✅ (file)	❌ — re-ingestable from <code>data/uploads/</code> PDFs
SQLite (<code>./smartfix.db</code>)	Machine parameters, shift logs, runtime config	✅ (file)	✅ via <code>scripts/backup_sqlite.sh</code>
JSONL (<code>./data/audit.jsonl</code>)	Append-only audit log	✅ (file)	⚠️ not in backup script — add manually
In-memory (Python lists/dicts)	Alerts, query log, snoozes, sessions, jobs	❌ — lost on restart	❌

ER diagram (SQLite)

machine_parameters			
TEXT	machine_id	PK	machine slug
TEXT	numeric_readings		JSON array of {key,label,unit,expected_min,expected_max}
TEXT	visual_checks		JSON array of {key,label,anomaly_when}
TEXT	updated_at		ISO timestamp

anomaly rules for

shift_logs			
INTEGER	id	PK	
TEXT	machine_id		FK-like; not enforced
TEXT	worker_label		derived server-side
TEXT	workstation_ip		
TEXT	readings		JSON dict of submitted values
TEXT	visual_checks		JSON dict of bools
TEXT	notes		free-form worker observation
TEXT	anomalies		JSON; frozen at submit time
INTEGER	severity		1-5
INTEGER	acknowledged		boolean 0/1
TEXT	phase		'start' or 'end'
TEXT	void_at		soft-delete timestamp (nullable)
TEXT	void_reason		required if void_at set
TEXT	voided_by		admin email
TEXT	created_at		ISO timestamp

app_config			
TEXT	key	PK	
TEXT	value		raw string; caller casts
TEXT	updated_at		

Tables — detail

machine_parameters

- One row per machine. Defines the admin-edited spec that EndShiftModal renders and that `compute_anomalies` uses to score worker submissions.
- `numeric_readings` JSON shape: `[{key, label, unit, expected_min, expected_max}]`.
- `visual_checks` JSON shape: `[{key, label, anomaly_when}]` — `anomaly_when=true` means "ticking the box is bad" (e.g. "Leaks observed"); `false` means "unticking is bad" (e.g. "Vibration normal").
- Seeded on first boot via `_DEFAULT_MACHINE_PARAMS` in `src/api.py` for the 4 demo machines. Existing rows are never overwritten by the seed.
- Schema lives in `src/store.py`; create + migrate is idempotent inside `init_store()`.

shift_logs

- Append-only. Insertion freezes `anomalies` + `severity` against the parameter spec that was in effect at submit time, so later edits to thresholds don't retroactively change history.
- Soft-delete via `void_at` (set by `POST /admin/shifts/{id}/void` with required reason). Voided rows stay queryable but are filtered out of `latest_shift_log()` so the HandoffBanner skips them.
- Index `idx_shift_logs_machine_created` on `(machine_id, created_at DESC)` for the recent-logs query.

- Migration paths (additive): `phase`, then `void_at` / `void_reason` / `voided_by`. SQLite has no `ADD COLUMN IF NOT EXISTS` so `init_store()` introspects `PRAGMA table_info` and conditionally `ALTER TABLE` s.

app_config

- Key/value store for admin-tunable runtime settings.
- Currently used keys:
 - `alert_threshold` (int, 1–25, default from `ALERT_THRESHOLD` env)
 - `alert_dedup_seconds` (int, 0–86400, default from `ALERT_DEDUP_SECONDS` env)
- Loaded into module-level globals at boot by `_load_runtime_config()` and re-loaded after every `PATCH /admin/config`. The DB row wins over `.env` when present.

ChromaDB

- **Path:** `./chroma_db/` (single SQLite-backed persistent client, see `src/db.py`).
- **Collection:** `machine_docs` with metric `cosine`.
- **Chunk shape** (one row per chunk):

Field	Source	Example
<code>id</code>	<code>{machine_id}_p{page}_c{chunk_idx}</code>	<code>IMM-750_p12_c03</code>
<code>embedding</code>	<code>all-MiniLM-L6-v2</code> (384-dim)	<code>[...]</code>
<code>document</code>	The chunk text	<code>"E-04 indicates oil pressure low..."</code>
<code>metadata.machine</code>	Used as the where-filter on every query	<code>INJECTION_MOLDING_MACHINE</code>
<code>metadata.document</code>	Original PDF filename	<code>IMM-750.pdf</code>
<code>metadata.page</code>	1-based page number	<code>12</code>
<code>metadata.section</code>	Heuristic section name	<code>troubleshooting</code>

- See [supplementary/CHUNKING_STRATEGY.md](#) for the ingestion algorithm (PDF → docling → chunk → embed → upsert).
- Re-ingestion (`POST /admin/machines/{id}/reingest`) first deletes all chunks where `metadata.machine == machine_id`, then re-runs ingestion against the archived `data/uploads/{machine_id}.pdf`.

data/audit.jsonl

- Append-only JSON-lines file. One event per line:

```
{"action": "machine.delete", "actor": "admin@tecdiva.local", "status": "success", "target": "OLD_MACHINE", "ip":
```

- Event categories: `auth.*`, `machine.*`, `shift.*`, `alert.*`, `config.*`.
- Read via `audit.read(limit, action_prefix)` (powers `GET /admin/audit`).
- **Not currently in the backup script** — add a tar of `data/audit.jsonl` alongside the SQLite backup.

In-memory state (`src/api.py`)

Everything below is **lost on server restart**. P1 hardening should move these to SQLite.

Variable	Shape	Purpose
<code>_alerts</code>	<code>list[dict]</code>	Fired alerts, newest at the end. Capped only by <code>DELETE /admin/alerts</code> .
<code>_alert_snoozes</code>	<code>{machine_id: ISO-until}</code>	Per-machine "be quiet" windows.
<code>_query_log</code>	<code>list[dict]</code>	Every <code>/query</code> call, used by <code>/admin/analytics</code> . FIFO-capped at <code>QUERY_LOG_MAX = 20000</code> .
<code>_machine_metadata</code>	<code>{machine_id: {display_name, category, significance, icon, description, suggested_questions}}</code>	Editable metadata that <code>_list_machines_basic</code> reads. Lost on restart — re-derivable from defaults + PDFs, but admin edits vanish.
<code>_worker_sessions</code>	<code>{cookie_value: {domain, machine_id?, workstation_ip?, created_at}}</code>	12-hour worker sessions.
<code>_admin_sessions</code>	<code>{cookie_value: {email, created_at}}</code>	30-day admin sessions.
<code>_magic_tokens</code>	<code>{token: {email, expires_at}}</code>	Single-use sign-in tokens. Popped on verify.
<code>_jobs</code>	<code>{job_id: {status, step, progress, ...}}</code>	Ingestion job state for the progress bar.

File layout cheat sheet

```
chroma_db/           # vector index – DON'T edit by hand
smartfix.db          # SQLite store (gitignored)
smartfix.db-shm      # SQLite WAL shared memory (gitignored)
smartfix.db-wal      # SQLite WAL (gitignored)
data/
  audit.jsonl        # append-only audit log
  uploads/           # archived machine PDFs
  uploads/icons/     # admin-uploaded custom machine icons
  workstations.json  # IP → machine bindings (manual edit, restart to reload)
```